

SteeringNET - A Deep-Learning based Obstacle Avoidance Approach for Autonomous Driving

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Abstract—In this paper, a deep-learning based model named SteeringNET is proposed for obstacles avoidance. Three categories of obstacles were chosen, namely, two-wheelers, cars, and buses. The Ego Vehicle (EGV) was defined using a kinematic model and integrated with the Adaptive Model Predictive Control (MPC) to simulate the obstacle avoidance strategies for human drivers. The dataset for the human driver obstacle avoidance trajectories was generated using MATLAB simulations. A total of 14040 trajectories: 5400, 2160 and 6480 for stationary two-wheeler, car, bus obstacles were obtained. Conventional MPC based models are computationally expensive for obstacle-free trajectory generation and are unsuitable for real-time applications. Our model readily predicts the throttle, steering angle values for negotiating different obstacles and the prediction time is significantly lesser than the Adaptive MPC. Four different versions of this model are implemented namely, Fully Connected Neural Network (FCNN), FCNN with custom loss, Long Short-Term Memory (LSTM) and LSTM with custom loss. The experimental results show that LSTM with custom loss algorithm outperforms the other three deep-learning models.

Index Terms—ADAS, Deep Neural Networks, Vehicle Motion Planning, Steering Intelligence, Obstacle Free Trajectories

I. INTRODUCTION

Artificial Intelligence (AI) plays major role in the development of Advanced Driver Assistance Systems (ADAS) to ensure road safety by incorporating aspects like driver attention monitoring, lane keeping, obstacle detection and avoidance, cruise control, etc. [1]. ADAS systems are commonly employed in passenger cars which aids the human-driver in critical driving situations by making timely interventions to avoid accidents [2]. A 2019 road accidents report by MoRTH, GoI states that most of the road accidents are caused by human errors [3]. Common mistakes in road crashes caused by drivers include sudden lane change without indication, traffic rule violation, over speed, miscalculation of other vehicles' velocity during overtaking process, etc. The challenges of human drivers in road safety under different contexts like low light conditions, adverse weather, etc. is explored by

researchers in the past [4]. The vehicle confronts numerous traffic obstacles which could become a possible hazard to safety when driver response time is insufficient to safely negotiate these obstacles.

An efficient drive-assist system must be able to perceive the environment and inform drivers about the presence of other vehicles, their mobility pattern, velocity, etc. for crash mitigation [5]. A detailed understanding of current motion pattern of the other vehicles is required so that the ego vehicle (EGV) can plan a safe trajectory. Supervised learning assisted maneuver-based vehicle motion prediction framework for dynamic environments using Long-short term memory (LSTM) based approach is investigated by several researchers [6]. On this front, we investigate different LSTM based approaches in this paper. Four versions of SteeringNET model is implemented and the predictions are compared against standard adaptive MPC based *obstacle-free* trajectory generation approach. The model is capable of predicting the required throttle and steering angles to avoid static obstacles.

This paper is organized as follows. Section 2 provides a concise review on the related works on vehicle dynamic models and deep learning approaches. Obstacle free trajectory computation with Adaptive MPC is explained in Section 3. Datasets and pre-processing methods used for training the proposed SteeringNET models are described in Section 4. The conceptual architecture of different proposed SteeringNET models along with the implementation is discussed in Section 5. Results and discussions are given in Section 6 and the conclusion follows in Section 7.

II. RELATED WORKS

Road environment perception, path planning and trajectory tracking are prerequisites for obstacle avoidance systems. Vehicle model influences the motion tracking and trajectory planning techniques. A trajectory tracking approach for a front-wheel driven kinematic vehicle model is proposed in [7]. An approach based on MPC is proposed in [8] to improve vehicle handling characteristics with independent wheel

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torque, side slip control. A bicycle model based nonlinear MPC cooperative trajectory planning is proposed in [9]. A path-planning algorithm proposed in [10] allows cooperation between autonomous and human-driven vehicles. A cooperative path-planning and tracking controller using a high fidelity vehicle model is proposed in [11] to generate obstacle avoidance maneuver. A consistent trajectory prediction approach using LSTM network is proposed in [12] to accurately predict future trajectories of other vehicles around it. Deep convolution neural network approach proposed in

[13] mimics human drivers' decisions such as steering angle, acceleration and deceleration. An approach for multi-modal trajectory prediction of surrounding vehicles is proposed in [14]. The vehicle trajectory prediction accuracy of LSTM based approach is significantly higher than the conventional trajectory prediction techniques [15]. A hierarchical planning architecture is proposed in [16] for obstacle avoidance. Transformer network and larger bidirectional transformer-based approach is proposed in [17] to predict the trajectories of the individual people. Deep reinforcement learning [18] have been utilized for their ability to adapt in complex traffic scenarios and are being employed in motion planning for the safety of autonomous vehicles. A deep learning based overtaking advice system is proposed in [19] to assist drivers to ensure road safety during overtaking process. A neural network assisted approximation-based adaptive nonlinear control is proposed in [20] to compensate the external disturbances and parametric uncertainties. A dynamic rollover index assisted optimal tilting approach is proposed in [21] to tilt vehicles when their rollover index go beyond the expected optimal threshold values.

From related works, it was observed that MPC-based approaches can be effectively used to generate vehicle trajectories to train deep learning models which can then be subsequently used for the real-time applications. In the next section we demonstrate an obstacle-free trajectory computation technique to generate datasets for training the SteeringNET model.

III. OBSTACLE FREE TRAJECTORY COMPUTATION WITH ADAPTIVE MPC

A. Modelling obstacle avoidance behaviour of human drivers

The proposed SteeringNET is intended to generate obstacle free trajectories for AD/ADAS, which requires a lot of benchmark dataset with forecasted steering angle and throttle values for training. In this research, Adaptive MPC simulation is used to generate obstacle free trajectories, which are considered to be the trajectories followed by ideal human drivers. The kinematic model is used in the Adaptive MPC to simulate the obstacle free trajectory as shown in Fig. 1.

Eq. (1) represents the non-linear kinematic model equations with global reference point on rear axle.

$$\begin{aligned} \dot{x} &= v \cos \Psi; & \dot{y} &= v \sin \Psi, \\ \dot{\Psi} &= \frac{v}{L} \tan \delta; & \dot{v} &= 0.5 \times T. \end{aligned} \quad (1)$$

where $\dot{x}$ is the rate of change of global X position and v is the longitudinal speed of the vehicle, $\dot{y}$ is the rate of change

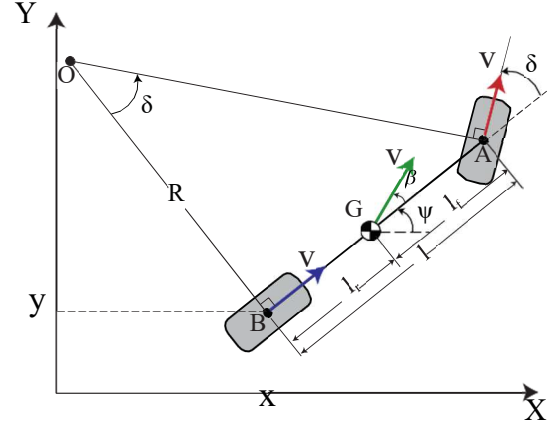


Fig. 1. Kinematic model used in obstacle avoidance strategy

of global Y position of the vehicle, $\dot{\Psi}$ is the yaw rate, i.e., the rate of change of heading angle of the vehicle (0 when facing east, counterclockwise positive), T is throttle (positive when accelerating, negative when decelerating). A linear throttle map has been assumed for simplicity.

The SteeringNET datasets are generated by the adaptive MPC based obstacle free trajectory generation technique for three different obstacle types. The initial velocity, distance between EGV and obstacle were kept unchanged. The architecture of adaptive MPC based obstacle avoidance strategy that was implemented in MATLAB environment is shown in Fig. 2.

The position, velocity and steering angle of the EGV are given as initial inputs to the Adaptive MPC. The constraint generator defines road environment along with lanes and obstacles. It assumes two lanes with 4.5m width for each. From these inputs, the Adaptive MPC predicts the throttle and steering angle for the EGV to move to the next position with the motivation of obstacle avoidance. Further the plant model generator uses the outcome of the Adaptive MPC as feedback to feed the new input values for Adaptive MPC. The obstacle detector module ensures whether the EGV is away from the collision threat or not. Hence the EGV can keep itself away from the obstacle by changing the lane. As the obstacle threat disappears, the EGV comes back to its lane. A scenario for the EGV to take an obstacle free trajectory using Adaptive MPC to avoid an obstacle is shown in Fig. 3.

The distance between the EGV and the obstacle at each instance is calculated. Based on the distance, the Adaptive MPC changes the lane for the EGV to ensure the obstacle avoidance. The path travelled by the EGV is recorded for datasets. The red colour dashed lines represents the safe zone around the obstacle. The green colour dashed line in Figure 3 denotes the slopes computed by the Adaptive MPC controller

at each point in the path. In order to train the SteeringNET, we considered 3 different obstacles at various distances and various velocities. The Adaptive MPC model outputs the throttle and steering angle values during obstacle avoidance. These values are collected to train the SteeringNET.

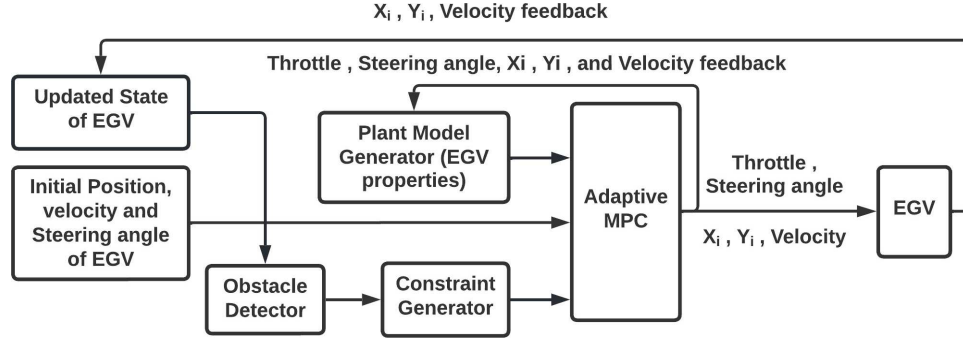


Fig. 2. Simulation model of Adaptive MPC controller for obstacle avoidance strategy

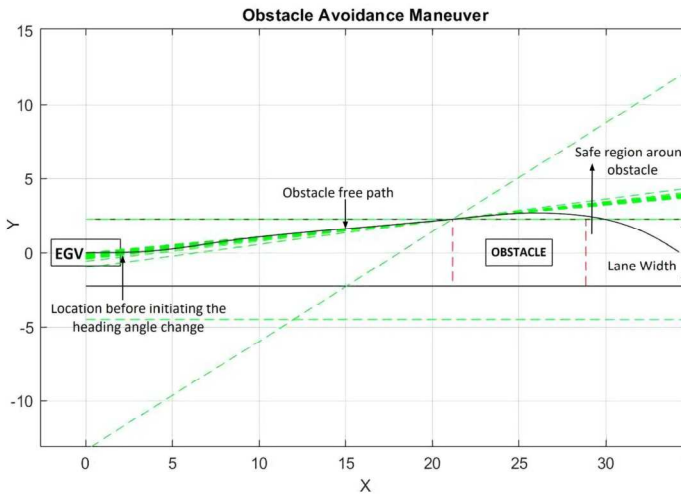


Fig. 3. Scenario of obstacle avoidance performed by EGV

B. SteeringNET Dataset Acquisition

The datasets are created for different obstacles at different distances and for a range of vehicle speeds. Using the adaptive of 2 input units followed by multiple hidden layers, which

MPC based obstacle avoidance strategy, the smooth path trajectories were generated, which include the steering and throttle outputs. These steering and throttle outputs were used to train the deep neural networks. A total of 14,040 trajectories were generated for two-wheeler (5400), four-wheeler (2160) and bus (6480) obstacles. The given dataset consists of two input parameters which are the current distance(m), current velocity(m/s) of the vehicle. The output parameters of the dataset consist of sixty-three points of throttle and steering angle values. The dataset split is 75% training, 25% validation.

IV. CONCEPTUAL ARCHITECTURE AND IMPLEMENTATION OF THE PROPOSED DEEP LEARNING BASED STEERINGNET MODELS

A deep neural network model based SteeringNET is proposed to forecast of the obstacle-free trajectory by considering the distance between EGV and obstacle and the current velocity of EGV. The base architecture of all the proposed models consists

passes their activation to the following final dense layer of 126 units, which outputs the model's predictions of sixty-three points of throttle and steering angle values. All the dense layers use the ReLU activation function and incorporate dropout layers of dropout rate 0.5. Kernel regularizers are employed to apply penalties on the last layer to avoid over fitting while training for longer epochs. Multiple models of similar architecture are trained for each type obstacle avoidance. The models are trained for 200-300 epochs with a batch size of 64. The Adam optimizer is used, which is initialized with a learning rate of 1e-3. Callbacks such as reduce learning rate on plateau and early stopping are applied while training to reach convergence quickly and avoid over fitting.

A. FCNN based SteeringNET model

The fully connected neural network used for developing SteeringNET consists of two input units, i.e., vehicle's current distance and velocity. The output of a particular layer in the fully connected layer is computed as shown in the Eq. (2).

$$f^{[l]}(x) = R\left(b^{[l]} + \sum_{i=1}^n w_i^{[l]} x_i\right) \quad (2)$$

$$R(z) = \max(0, z),$$

where $f^{[l]}$ represents activation outputs from l^{th} layer, x is the input, $w^{[l]}$ is the weight matrix, $b^{[l]}$ is the bias vector in the l^{th} layer and $R(z)$ denotes the ReLU activation function. The vector output is reshaped to obtain 63 future instances of throttle and steering angle respectively.

B. FCNN based SteeringNET with Custom Loss Function

The initial FCNN model is trained using standard mean square error (MSE) loss function. The model performance is further enhanced by training using a custom loss function which is a residual sum of squares SS_{res} (see Eq. (3)) representing the cumulative sum of squares of residual loss from each output data point.

$$SS_{res} = \sum_{i=1}^n (x_i - \hat{x}_i)^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (3)$$

where x_i is true throttle value, $\hat{x}_i$ is the predicted throttle value, y_i is the true steering angle and $\hat{y}_i$ is the predicted steering angle, n is the total number of predicting outputs.

C. LSTM network based SteeringNET model

A recurrent neural network (RNN) based approach has been taken to improve the model performance since the SteeringNET output predictions are basically discrete time series data i.e. trajectory values at continuous time intervals. LSTMs are capable of consistently predicting accurate future longitudinal telemetry i.e. throttle and lateral i.e. steering angle trajectories for vehicles to avoid the obstacles. The equations governing the output of each LSTM cell is denoted mathematically as follows in Eq. (4).

$$\begin{aligned} i_t &= \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \\ f_t &= \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \\ o_t &= \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_{t-1} + b_o) \\ c_t &= f_t c_{t-1} + i_t \tanh(W_{wc}x_t + W_{hc}h_{t-1} + b_c) \\ h_t &= o_t \tanh(c_t), \end{aligned} \quad (4)$$

where i_t , f_t , o_t are the outputs from the input gate, forget gate and output gate respectively, x_t is input at current time step, h_{t-1} is the output from previous time step, W is weight parameter matrix, b is bias, c being the cell state and σ , $\tanh$ denotes sigmoid activation function, hyperbolic tangent function respectively.

D. LSTM network based SteeringNET model with Custom Loss Function

The model performance of LSTM based network is also improved similar to FCNN with custom loss by training the model using SS_{res} custom loss function which has been previously trained using MSE loss.

V. RESULTS AND DISCUSSION

The parameters used to generate obstacle free trajectories are given in Table I. Scenarios considered for generating the obstacle free trajectories are reported in Table II.

TABLE I
SIMULATION PARAMETERS FOR ADAPTIVE MPC BASED OBSTACLE FREE TRAJECTORY GENERATION

Parameters	Values
Obstacle type	Bus / Four-wheeler / Two-wheeler
Dimension: Bus	Length: 14m, Width: 2.5m
Dimension: Car	Length: 3.84m; Width: 1.74m
Dimension: 2-wheeler	Length: 2.30m; Width: 0.88m
EGV dimension	Length: 3.84m; Width: 1.74m
EGV velocity	Driver dependence (range) m/s
Initial steer & lane	0 degree, Right
Width & no. of lanes	4.5m, Two

The vehicle trajectory, heading angle and velocity from its starting location to end location during obstacle avoidance process is shown in Fig. 4(a)-(c) respectively. The throttle and steering angle inputs generated by the algorithm to observe these trajectories are shown in Figure 4(d)-(e).

TABLE II
FEASIBLE USE CASES

Obstacle type	Distance between EGV and obstacle range (m)	Initial velocity range (m/s)
Two wheeler	5-7, 8-10 10-12, 12-14 14-16, 16-18	10-20, 20-50 20-80, 50-90 60-120, 80-120
Four wheeler	7-9, 8-10 9-11, 10-12 12-14, 14-16	30-40, 40-50 50-70, 70-80 70-120, 80-100
Bus	6-8, 8-10 10-12, 12-14 14-16, 16-18	30-40, 30-40 30-80, 40-90 40-90, 50-100

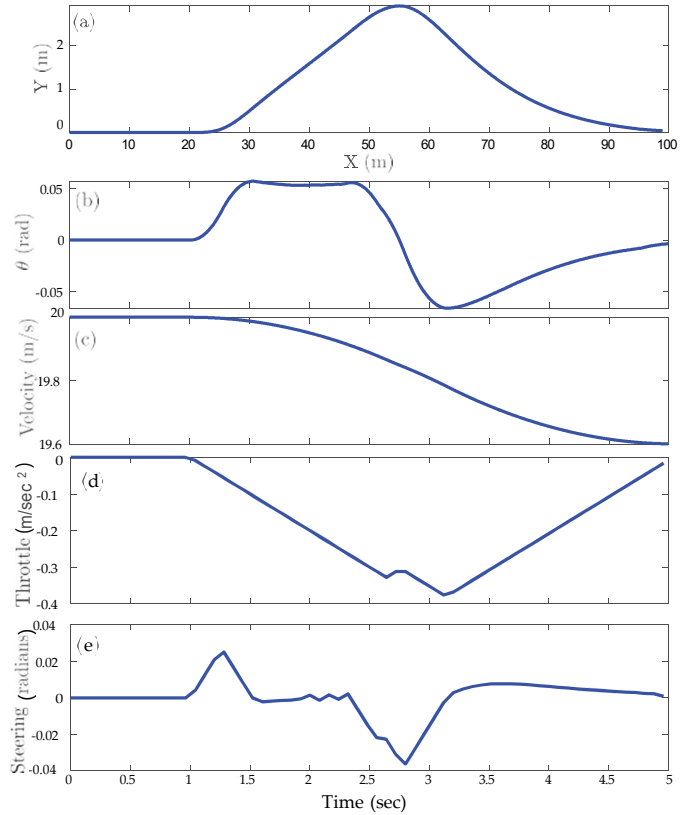


Fig. 4. Results: (a) Reference Position; (b) Heading; (c) Velocity; (d) Throttle; (e) Steering

The results obtained using proposed models were compared using different evaluation metrics, namely, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Explained Variance Score (EVS). MAE is a metric to measure the model's absolute deviation. RMSE is also a metric to measure error. It is the square root of mean of squared differences between the predicted and true values. EVS measures the proportion of variance. Fig. 5. plots SteeringNET predictions and ground truth values for the obstacles two-wheeler, four-wheeler and bus. Results obtained from evaluating each of these models for different obstacles on the testing data are summarized in Table III. It achieved an overall explained variance score of 0.99 for two-wheeler, 0.93 for four-wheeler obstacle and 0.98 for bus obstacle. Also, it is quite evident that LSTMs based neural network performed much better than fully connected network owing to the sequential nature of data.

TABLE III
RESULTS OBTAINED FROM TWO-WHEELER, FOUR-WHEELER AND BUS OBSTACLE AVOIDANCE MODELS RESPECTIVELY

Model	MAE	RMSE	Throttle EVS	Steering Angle EVS	Overall EVS
Neural Network	0.12, 0.09, 0.10	0.17, 0.15, 0.16	0.39, 0.73, 0.70	0.62, 0.70, 0.57	0.73, 0.75, 0.76
Neural Network (SS _{res} Loss)	0.10, 0.09, 0.09	0.16, 0.14, 0.15	0.49, 0.80, 0.79	0.67, 0.70, 0.62	0.78, 0.78, 0.79
LSTM Network	0.04, 0.06, 0.04	0.08, 0.09, 0.07	0.90, 0.86, 0.95	0.91, 0.90, 0.91	0.95, 0.90, 0.95
LSTM Network (SS _{res} Loss)	0.02, 0.05, 0.02	0.03, 0.08, 0.04	0.98, 0.89, 0.97	0.98, 0.94, 0.97	0.99, 0.93, 0.98

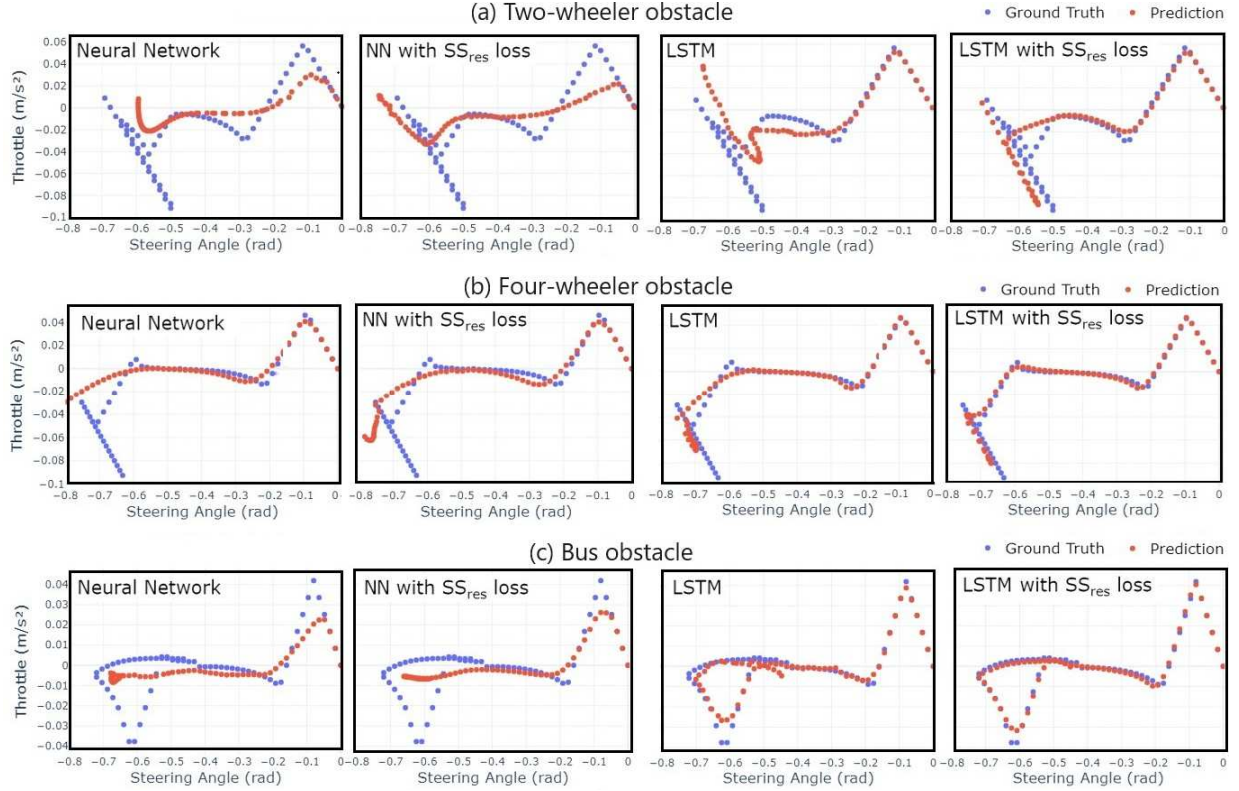


Fig. 5. SteeringNET predictions versus ground truth values for different obstacles: (a) Two-wheeler; (b) Four-wheeler; (c) Bus

that LSTM cells are leveraging on. Notably, the use of SS_{res} demonstrates a formidable improvement in all the model's performance by a considerable margin. Model performance for 2-wheeler is slightly better than 4-wheeler and bus obstacles.

From prediction plots of each obstacle, we could clearly interpret that the LSTM network models achieve solid results and are pretty close to the ground truth with minimum error. The performance is amplified using the SS_{res} custom loss function that further diminished the error to develop a greater fit on the data.

Table IV shows the inference time comparison of different models for the prediction of obstacle avoidance trajectory around three types of static objects two-wheeler, four-wheeler, bus. The inference time is measured considering hundred random samples from each obstacle type. The input parameters in the random samples vary as follows. In the two-wheeler obstacle samples, distance varies between 11.1 - 49.7(m), and velocity varies between 5.1 - 11.9(m/s). In the four-wheeler obstacle samples, distance varies between 19.3 - 39.8(m), and velocity varies between 6.0 - 8.9(m/s). In the bus obstacle samples, distance varies between 21.1 - 60.8(m), and velocity varies between 6.0 - 11.9(m/s). From the measured observations, it is obvious that the deep learning models are incredibly fast, with nearly 3200% reduction on average in

prediction time while using deep learning models compared to MPC. We also observe that LSTM based networks take 39% more time on average than fully connected neural networks owing to their sequential nature.

TABLE IV
EXECUTION TIME (IN MILLI SECONDS) COMPARISON FOR DIFFERENT OBSTACLES USING DIFFERENT MODELS

Obstacle	MPC	NN	NN (with loss)	LSTM	LSTM (with loss)
2-Wheeler	2555.9	59.1	59.0	83.9	83.5
4-Wheeler	2538.8	61.3	62.3	82.3	86.5
Bus	2194.4	59.2	59.4	83.0	82.7

VI. CONCLUSION

In this paper, four architectures of neural-network based SteeringNET model (FCNN, FCNN with custom loss, LSTM and LSTM with custom loss) are proposed to predict vehicle trajectory to avoid three different types of static obstacles. The models are capable of bypassing the computational time required to generate obstacle free trajectories around static obstacles to provide quick and spontaneous throttle, steering angle values to accurately predict static obstacle avoidance trajectories. All models that incorporated LSTM cells have

persistently shown better results than FCNNs with an average reduction of 56% in RMSE loss and an increase in 24% overall variance score. Also, adopting a custom loss function SS_{res} to train these models has been shown to improve the models notable with an average reduction in RMSE loss by 19% and increase in overall variance score by 4% across all the models. From the results acquired, it is quite evident that Long Short-Term Memory (LSTM) network based deep learning predictive models have outperformed the other conventional models. Finally, LSTM based SteeringNET model trained using SS_{res} custom loss scored 0.52 whereas FCNN scored a 0.161 in average RMSE loss, which is a 67% reduction. The average explained variance being 0.96 and 0.74 for LSTM with custom loss and FCNN models respectively, which is a 29% rise in performance. From all the results and performance metrics, it could be safe to conclude that the proposed SteeringNET model using the LSTM architecture with SS_{res} custom loss yielded significant performance improvements proving that the model is capable of consistently predicting the accurate and optimal obstacle avoidance trajectory for different types of obstacles.

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